If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.

b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.

c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.

d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).

e. Bonus problems are not required for credit, but they are worth extra credit.

Definitions. The following definitions will be used in the problems.

Definition 1 (1-Nearest Neighbor (1-NN) Classifier). Given a set of labeled points $\{(x_i, y_i)\}$ with $x_i \in \mathbb{R}^d$, the *1-nearest neighbor classifier* assigns a label to any query point $x \in \mathbb{R}^d$ by finding the closest training point in Euclidean distance:

$$\hat{y}(x) = y_j \quad \text{where } j = \arg\min_i \|x - x_i\|.$$

Definition 2 (Perpendicular Bisector Hyperplane). Given two points $a, b \in \mathbb{R}^d$, the perpendicular bisector hyperplane is the set of all points equidistant from a and b:

$$H = \{x \in \mathbb{R}^d : \|x - a\| = \|x - b\|\}.$$

Geometrically, H is the $(d-1)$-dimensional hyperplane that is perpendicular to the line segment joining a and b, and passes through the midpoint $\frac{a+b}{2}$.

Problems

Problem 1. (Nearest Neighbor Geometry.) Let $x \in \mathbb{R}^d$ and consider three points $a, b, c \in \mathbb{R}^d$.

(a) Show that a is closer to x than b if and only if

$$(x - \tfrac{a+b}{2})^\top (a - b) > 0.$$

(b) Conclude that the 1-NN decision boundary between a and b is the perpendicular bisector hyperplane.

Solution.
(a) Compare squared distances:

$$
\begin{aligned}
0 > \|x-a\|^2 - \|x-b\|^2 &= (x-a)^\top(x-a) - (x-b)^\top(x-b) \\
&= -2x^\top(a-b) + (a^\top a - b^\top b) \\
&= -2(x - \tfrac{a+b}{2})^\top(a-b).
\end{aligned}
$$

Rewriting, Thus $\|x-a\|^2 < \|x-b\|^2 \iff (x - \frac{a+b}{2})^\top(a-b) > 0$.

(b) The set where distances are equal is given by

$$(x - \tfrac{a+b}{2})^\top(a-b) = 0,$$

which is the equation of the perpendicular bisector hyperplane between a and b.

Problem 2. (Weighted Norm Decision Boundaries.) In weighted 1-nearest neighbor (1-NN) classification, the decision boundary between two training points $a, b \in \mathbb{R}^d$ is defined as the set of points x such that

$$d_W(x,a) = d_W(x,b),$$

where the weighted Euclidean distance is

$$d_W(x,y) = \sqrt{(x-y)^\top W(x-y)}, \qquad W = \mathrm{diag}(\alpha_1,\ldots,\alpha_d) = \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_d \end{bmatrix}, \ \alpha_i > 0.$$

(a) Show that the decision boundary has the form of a hyperplane

$$c^\top x = d,$$

for some vector c and scalar d depending on a, b, W.

(b) Specialize your result to $d = 2$ and write the equation of the decision boundary in terms of $\mathbb{R}^2$coordinates x_1, x_2.

(c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.

(d) Again with $d = 2$, let $z = Sx$ with $S = \mathrm{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2})$. Show that in the z-coordinates the decision boundary is the usual perpendicular bisector of the transformed points Sa and Sb. Explain how the parameters (α_1, α_2) rescale the coordinate axes: if $\alpha_1 > \alpha_2$, distances along the x_1 direction are stretched more, so the first feature has greater influence in the weighted 1-NN classification, and similarly for $\alpha_2 > \alpha_1$.

Solution.

(a) Boundary equation as hyperplane. By definition we have that

$$d_W(x,a) = d_W(x,b) \iff (x-a)^\top W(x-a) = (x-b)^\top W(x-b).$$

Expanding and canceling $x^\top Wx$ from both sides, we have that

$$-2x^\top Wa + a^\top Wa \;=\; -2x^\top Wb + b^\top Wb \iff x^\top W(b-a) \;=\; \tfrac{1}{2}\Big(b^\top Wb - a^\top Wa\Big).$$

Equivalently,

$$\boxed{(W(b-a))^\top x \;=\; \tfrac{1}{2}\big(\|b\|_W^2 - \|a\|_W^2\big), \quad \|u\|_W^2 := u^\top Wu.}$$

This is an affine hyperplane of the given form with $c = W(b-a)$, and $d = \frac{1}{2}\big(\|b\|_W^2 - \|a\|_W^2\big)$. Note that we call $c = W(b-a)$ the **normal vector** for this hyperplane.

(b) Boundary equation in $d = 2$. Specializing to $d = 2$, we have

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \quad W = \mathbf{diag}(\alpha_1, \alpha_2).$$

Then

$$W(b-a) = \begin{bmatrix} \alpha_1(b_1 - a_1) \\ \alpha_2(b_2 - a_2) \end{bmatrix},$$

and

$$b^\top Wb - a^\top Wa = \alpha_1(b_1^2 - a_1^2) + \alpha_2(b_2^2 - a_2^2).$$

Substituting into the general formula yields

$$\boxed{\alpha_1(b_1 - a_1)x_1 + \alpha_2(b_2 - a_2)x_2 = \tfrac{1}{2}\big[\alpha_1(b_1^2 - a_1^2) + \alpha_2(b_2^2 - a_2^2)\big].}$$

Equivalently, using $b_i^2 - a_i^2 = (b_i - a_i)(b_i + a_i)$,

$$\alpha_1(b_1 - a_1)x_1 + \alpha_2(b_2 - a_2)x_2 = \tfrac{1}{2}[\alpha_1(b_1 - a_1)(b_1 + a_1) + \alpha_2(b_2 - a_2)(b_2 + a_2)].$$

Hence, the decision boundary is a straight line in $\mathbb{R}^2$. Further, if we bring the right-hand side to the left and group by $(b_i - a_i)$ we get that

$$\alpha_1(b_1 - a_1)\Big(x_1 - \tfrac{a_1 + b_1}{2}\Big) + \alpha_2(b_2 - a_2)\Big(x_2 - \tfrac{a_2 + b_2}{2}\Big) = 0.$$

so this is just

$$(b-a)^\top W\left(x - \frac{b+a}{2}\right) = 0$$

Indeed, this is the weighted perpendicular bisector hyperplane between a and b à la **Problem 1**.

(c) Perpendicularity to the segment ab.
We claim that for $d = 2$ the decision boundary

$$(b-a)^\top W\big(x - \tfrac{a+b}{2}\big) = 0$$

is not, in general, perpendicular to the segment ab in the ordinary (Euclidean) coordinates when $\alpha_1 \neq \alpha_2$.
To see this is the case, start with the following observation. The Euclidean normal to the segment ab is the vector $b - a$. The Euclidean normal to the boundary

$$\{x : (b-a)^\top W(x - \tfrac{a+b}{2}) = 0\}$$

is the vector

$$n \;=\; W(b-a) \;=\; \begin{pmatrix} \alpha_1(b_1-a_1) \\ \alpha_2(b_2-a_2) \end{pmatrix}.$$

The boundary is perpendicular to the segment ab in the usual Euclidean sense precisely when the two normals are parallel, i.e. when there exists a scalar $\lambda \neq 0$ with

$$W(b-a) = \lambda(b-a).$$

Writing this componentwise gives

$$\alpha_1(b_1-a_1) = \lambda(b_1-a_1), \qquad \alpha_2(b_2-a_2) = \lambda(b_2-a_2).$$

If both differences $b_1 - a_1$ and $b_2 - a_2$ are nonzero, we can divide and obtain $\alpha_1 = \lambda = \alpha_2$, so

$$\alpha_1 = \alpha_2$$

is necessary for parallelism. Thus when $\alpha_1 \neq \alpha_2$ (and $b_1 \neq a_1$, $b_2 \neq a_2$) the vector $W(b-a)$ is not parallel to $b-a$, hence the boundary is *not* perpendicular to the segment ab in the usual coordinates.

Remark (degenerate cases). If one of the coordinate differences vanishes, say $b_1 = a_1$, then $b-a$ is parallel to the second coordinate axis and

$$W(b-a) = \alpha_2(b_2-a_2)e_2$$

is parallel to $b-a$ regardless of α_1, α_2. In that special (axis-aligned) case the boundary is perpendicular to ab even when $\alpha_1 \neq \alpha_2$.
Alternative/shorter (but valid) argument. In the standard Euclidean case ($W = I$), the normal is $(b-a)$, so the boundary is the *perpendicular bisector* of ab. Here the normal is $W(b-a)$, so the boundary is perpendicular to ab if and only if $W(b-a)$ is parallel to $(b-a)$, i.e.

$$W(b-a) = \lambda\,(b-a) \quad \text{for some } \lambda \in \mathbb{R}.$$

With $W = \operatorname{diag}(\alpha, \beta)$ this occurs when either $\alpha = \beta$ (isotropic scaling) *or* $(b-a)$ is aligned with a coordinate axis (so that $(b-a)$ is an eigenvector of W). Thus, for generic a, b and $\alpha \neq \beta$, the boundary is *not* perpendicular to ab.

(d) Feature-scaling interpretation.
Let $d = 2$ and define

$$S = \mathbf{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2}), \qquad z = Sx.$$

Note that $W = S^\top S = S^2$. For any $x, a \in \mathbb{R}^2$ we have

$$d_W(x,a) = \sqrt{(x-a)^\top W(x-a)} = \sqrt{(x-a)^\top S^\top S(x-a)} = \|S(x-a)\|_2 = \|z - Sa\|_2,$$

where $\|\cdot\|_2$ is the usual Euclidean norm and $z = Sx$.
Thus the weighted boundary condition $d_W(x,a) = d_W(x,b)$ becomes in z-coordinates

$$\|z - Sa\|_2 = \|z - Sb\|_2.$$

Squaring and simplifying yields the standard perpendicular-bisector equation in z-space:

$$(Sb - Sa)^\top\Big(z - \tfrac{Sa+Sb}{2}\Big) = 0,$$

so the decision boundary is the (Euclidean) perpendicular bisector of the points Sa and Sb in the z-coordinates.
Rescaling/geometric intuition. The linear map $S = \mathbf{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2})$ rescales the coordinate axes:

$$z_1 = \sqrt{\alpha_1}\, x_1, \qquad z_2 = \sqrt{\alpha_2}\, x_2.$$

If $\alpha_1 > \alpha_2$ then differences along the x_1-axis are amplified more in z-space than differences along x_2. Equivalently, the metric induced by W stretches distances in the x_1-direction relative to the x_2-direction, so the first feature has greater influence on nearest-neighbor decisions. (If $\alpha_2 > \alpha_1$ the reverse holds.)

Problem 3. (Norm Effects on k-Mean.)
Consider two clusters of points in $\mathbb{R}^2$: $A = \{(0,0), (0,2)\}$ and $B = \{(5,0), (5,2)\}$.

(a) Compute the centroids of A and B.

(b) For a new point $x = (2,1)$, compute its distance to both centroids in the ℓ_2 and ℓ_1 norms.

(c) Show that the assignment of x can differ depending on the choice of norm.

Solution.
(a) Centroids are

$$c_A = \frac{1}{2}\left((0,0),(0,2)\right) = (0,1), \quad c_B = \frac{1}{2}\left((5,0),(5,2)\right) = (5,1).$$

(b) Distances from $x = (2,1)$:

$$\|x - c_A\|_2 = \sqrt{(2-0)^2 + (1-1)^2} = 2, \quad \|x - c_B\|_2 = \sqrt{(2-5)^2 + (1-1)^2} = 3.$$

$$\|x - c_A\|_1 = |2-0| + |1-1| = 2, \quad \|x - c_B\|_1 = |2-5| + |1-1| = 3.$$

Here both agree. But if we move x to $(2,5)$:

$$\|x - c_A\|_2 = \sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.47, \quad \|x - c_B\|_2 = \sqrt{(-3)^2 + 4^2} = 5.$$

So Euclidean assigns to A. In ℓ_1:

$$\|x - c_A\|_1 = |2| + |4| = 6, \quad \|x - c_B\|_1 = |-3| + |4| = 7.$$

Still A. For some placements, however, ℓ_1 and ℓ_2 can disagree (exercise: try $x = (3,10)$).
(c) Hence, the cluster assignment depends on the geometry induced by the chosen norm. Different norms stretch space differently, which shifts decision boundaries.

Problem 4. (1-NN Decision Boundaries under Different Norms.) Consider two training points $a = (0,0)$ and $b = (2,1)$ with different labels.

(a) Under the ℓ_2 norm, write the equation of the 1-NN decision boundary (the set of x equidistant to a and b).

(b) Under the ℓ_1 norm, write the equation of the 1-NN decision boundary. (Hint: solve $|x_1| + |x_2| = |x_1 - 2| + |x_2 - 1|$ by cases on the signs of the absolute values.)

(c) Compare the two boundaries and explain the geometric difference.

Solution.
(a) ℓ_2 **norm.** The decision boundary consists of points equidistant from a and b under ℓ_2:

$$\|x-a\|_2^2 = \|x-b\|_2^2.$$

Expanding,

$$x^\top x - 2a^\top x + \|a\|^2 = x^\top x - 2b^\top x + \|b\|^2.$$

The $x^\top x$ terms cancel, giving

$$2(b-a)^\top x = \|b\|^2 - \|a\|^2.$$

Substituting $a=(0,0), b=(2,1)$:

$$2x_1 + x_2 = \tfrac{5}{2}.$$

Thus the ℓ_2 decision boundary is the straight line

$$\boxed{2x_1 + x_2 = 2.5}.$$

(b) ℓ_1 **norm.** Now solve

$$|x_1| + |x_2| = |x_1 - 2| + |x_2 - 1|.$$

Consider cases depending on the signs of (x_1, x_2) relative to $(2,1)$:
- For $0 \le x_1 \le 2,\ 0 \le x_2 \le 1$:

$$x_1 + x_2 = (2 - x_1) + (1 - x_2) \implies 2x_1 + 2x_2 = 3.$$

So the boundary is the segment

$$\boxed{x_2 = \tfrac{3}{2} - x_1, \quad 0.5 \le x_1 \le 1.5}.$$

- For $0 \le x_1 \le 2,\ x_2 \ge 1$:

$$x_1 + x_2 = (2 - x_1) + (x_2 - 1) \implies 2x_1 = 1.$$

So the boundary is the vertical ray

$$\boxed{x_1 = \tfrac{1}{2},\ \ x_2 \ge 1}.$$

- For $0 \le x_1 \le 2,\ x_2 \le 0$:

$$x_1 - x_2 = (2 - x_1) + (1 - x_2) \implies 2x_1 = 3.$$

So the boundary is the vertical ray

$$\boxed{x_1 = \tfrac{3}{2},\ \ x_2 \le 0}.$$

Other orthants give no solutions.

(c) Comparison.

- Under ℓ_2, the decision boundary is a single straight line (the perpendicular bisector of a and b).

- Under ℓ_1, the decision boundary is *piecewise linear*: a line segment plus two vertical rays, reflecting the diamond geometry of ℓ_1 balls.

Thus different norms produce qualitatively different 1-NN decision boundaries.

Problem 5. (Monotonicity of k-Means Objective.) Show that updating centroids as the mean reduces the objective.

Solution. There are a couple ways to solve this problem. First, we can solve it by using similar analysis to Problem 8 or alternatively, we can use derivatives from calculus (i.e. definition of decreasing functions).

Method 1: Let a current clustering be $C_1, \dots, C_k$ with sizes $m_j = |C_j|$ and current centroids $c_j \in \mathbb{R}^d$. The k-means objective is

$$J \;=\; \sum_{j=1}^{k} \sum_{x \in C_j} \|x - c_j\|^2.$$

Fix any cluster C_j and let its (data) mean be

$$\mu_j \;=\; \frac{1}{m_j} \sum_{x \in C_j} x.$$

We use the *variance decomposition identity* (a simple completing-the-square argument):

$$\sum_{x \in C_j} \|x - c\|^2 \;=\; \sum_{x \in C_j} \|x - \mu_j\|^2 \;+\; m_j \,\|c - \mu_j\|^2, \quad \text{for any } c \in \mathbb{R}^d. \tag{$*$}$$

Proof of $(*)$*:* Write $x - c = (x - \mu_j) + (\mu_j - c)$ and expand:

$$\|x - c\|^2 = \|x - \mu_j\|^2 + 2(x - \mu_j)^\top(\mu_j - c) + \|\mu_j - c\|^2.$$

Summing over $x \in C_j$ kills the cross term because $\sum_{x \in C_j}(x - \mu_j) = 0$, yielding $(*)$.

Apply $(*)$ twice to the current centroid c_j and to the updated centroid μ_j:

$$\sum_{x \in C_j} \|x - c_j\|^2 = \sum_{x \in C_j} \|x - \mu_j\|^2 \;+\; m_j \|c_j - \mu_j\|^2,$$

$$\sum_{x \in C_j} \|x - \mu_j\|^2 = \sum_{x \in C_j} \|x - \mu_j\|^2 \;+\; m_j \|\mu_j - \mu_j\|^2.$$

Therefore, the *change in the objective contributed by cluster* j when we update the centroid from c_j to μ_j is

$$\Delta J_j = \sum_{x \in C_j} \|x - \mu_j\|^2 \;-\; \sum_{x \in C_j} \|x - c_j\|^2 = -\, m_j \,\|c_j - \mu_j\|^2 \;\le\; 0,$$

with equality iff $c_j = \mu_j$.
Summing over $j = 1, \dots, k$ gives the global change

$$\Delta J \;=\; \sum_{j=1}^{k} \Delta J_j \;=\; -\sum_{j=1}^{k} m_j \,\|c_j - \mu_j\|^2 \;\le\; 0,$$

and strict decrease unless every centroid already equals its cluster mean.
In conclusion, the centroid *update step* of Lloyd's algorithm never increases the objective and strictly decreases it unless centroids are already at the cluster means. This argument is purely algebraic and does not invoke convexity.

(For completeness: we can also show this result for the assignment step.) Given fixed centroids $\{c_j\}$, reassigning each x to its *nearest* centroid replaces $\min_j \|x - c_j\|^2$ by its minimal value among the same set, so $\sum_i \min_j \|x_i - c_j\|^2$ cannot increase. Hence each full Lloyd iteration (assignment then update) produces a non-increasing sequence of J.

Method 2: Via Calculus. (Optional) Let $A = \{x_1, \ldots, x_m\}$ be assigned to one cluster with centroid c. The objective contribution is

$$J(c) = \sum_{i=1}^{m} \|x_i - c\|^2.$$

Expanding:

$$J(c) = \sum_i \left(\|x_i\|^2 - 2c^\top x_i + \|c\|^2\right).$$

The derivative with respect to c is

$$\nabla_c J(c) = -2\sum_i x_i + 2mc.$$

Now we know that for a function f, if $f'(x) < 0$ on an open interval (e..g, (a, b)), then f is decreasing on the interval. This is easy to check when its scalar. For the vector case, we can either compute the second derivative or we can check the inner product analogously to the condition in the scalar case.
Let $\bar{x} := \frac{1}{m}\sum_{i=1}^{m} x_i$. The gradient of the function is

$$\nabla_c J(c) = -2\sum_i x_i + 2mc = 2m(c - \bar{x}).$$

Indeed, in the scalar case $c \in \mathbb{R}$ we have that

$$J'(c) = 2m(c - \bar{x}).$$

Hence,

$$\begin{cases} J'(c) < 0 & \iff c < \bar{x} \quad \text{(function decreasing)} \\ J'(c) > 0 & \iff c > \bar{x} \quad \text{(function increasing)} \\ J'(c) = 0 & \iff c = \bar{x} \quad \text{(stationary point).} \end{cases}$$

Now in the vector case $c \in \mathbb{R}^d$, the function J decreases in a (small) step from c along direction d if the *directional derivative* is negative:

$$DJ(c; d) = \nabla J(c)^\top d = 2m(c - \bar{x})^\top d < 0.$$

A guaranteed decreasing direction is *toward the mean*, i.e. take $d = \bar{x} - c$. Then,

$$\nabla J(c)^\top (\bar{x} - c) = 2m(c - \bar{x})^\top (\bar{x} - c) = -2m\|c - \bar{x}\|^2 < 0,$$

unless $c = \bar{x}$.

The alternative to this check is "convexity" or the second derivative: indeed, since

$$\nabla_c^2 J(c) = 2mI \succ 0,$$

the function J is convex and has a unique minimizer at

$$c^\star = \bar{x}.$$

Thus, J decreases whenever c moves toward $\bar{x}$ and increases otherwise.
For posterity, here is the definition of convexity and connection to strictly decreasing functions.

Definition 3. A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is said to be *convex* if for all $x, y \in \mathbb{R}^d$,

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

Equivalently, f lies above its tangent plane at every point.

If f is twice differentiable, this is equivalent to requiring that the *Hessian* is positive semidefinite:

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x.$$

If $\nabla^2 f(x) \succ 0$, then f is *strictly convex*, and has a unique minimizer.
For a convex function, the gradient $\nabla f(x)$ always points in a direction of *increase*. Therefore, moving in the opposite direction of the gradient (i.e. $-\nabla f(x)$) locally decreases the function value. Formally, for a small step size $\eta > 0$,

$$f(x - \eta \nabla f(x)) \;\leq\; f(x) - \eta \|\nabla f(x)\|^2 + o(\eta),$$

which shows that f decreases whenever $\nabla f(x) \neq 0$.
Coming back to our case, we have that

$$\nabla_c J(c) = 2m(c - \bar{x}), \qquad \nabla_c^2 J(c) = 2mI \succ 0,$$

so J is *strictly convex*. Hence, J decreases whenever c moves *toward* the minimizer $\bar{x}$, and increases when c moves away from it.

Problem 6. (Matrix View of k-Means.) For a matrix $A \in \mathbb{R}^{m \times n}$ with entries a_{ij}, the **Frobenius norm** is defined as

$$\|A\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2}.$$

Equivalently, it is the Euclidean (ℓ_2) norm of the matrix viewed as a long vector of entries. Another useful identity is

$$\|A\|_F^2 = \operatorname{trace}(A^\top A) \quad \text{where } \operatorname{trace}(M) = \sum_{i=1}^{n} m_{ii} \text{ for } M \in \mathbb{R}^{n \times n}.$$

Let $X \in \mathbb{R}^{n \times d}$ be the data matrix whose ith row is $x_i^\top$. Let $H \in \{0, 1\}^{n \times k}$ be the assignment matrix with exactly one 1 per row (i.e., row i has a 1 in column $r(i)$ if x_i is assigned to cluster $r(i)$). Let $C \in \mathbb{R}^{k \times d}$ be the centroid matrix whose jth row is $c_j^\top$. Show carefully that the k-means objective with fixed assignments H equals

$$\sum_{i=1}^{n} \|x_i - c_{r(i)}\|_2^2 \;=\; \|X - HC\|_F^2.$$

(Provide both a row-wise argument and, if you like, a trace/Frobenius argument.)

Solution.

Row-wise argument. The ith row of HC is

$$(HC)_{i\cdot} \;=\; \sum_{j=1}^{k} h_{ij} c_j^\top \;=\; c_{r(i)}^\top,$$

since $h_{i\,r(i)} = 1$ and all other $h_{ij} = 0$. Therefore

$$\|X - HC\|_F^2 = \sum_{i=1}^{n} \left\| x_i^\top - (HC)_{i\cdot} \right\|_2^2 = \sum_{i=1}^{n} \| x_i - c_{r(i)} \|_2^2,$$

which is exactly the k-means cost with assignments H.

Trace/Frobenius argument (optional but instructive). Using $\|A\|_F^2 = \mathrm{trace}(A^\top A)$,

$$\|X - HC\|_F^2 = \mathrm{trace}\big((X-HC)^\top (X-HC)\big) = \mathrm{trace}(X^\top X) - 2\,\mathrm{trace}(C^\top H^\top X) + \mathrm{trace}(C^\top H^\top HC).$$

Here $H^\top H = \mathrm{diag}(m_1, \ldots, m_k)$ with $m_j = |C_j|$, and $(H^\top X)_{j\cdot} = \sum_{x_i \in C_j} x_i^\top$ collects cluster sums. Expanding the last term, we have that

$$\mathrm{trace}(C^\top H^\top HC) = \sum_{j=1}^{k} m_j \|c_j\|_2^2, \quad \text{and} \quad \mathrm{trace}(C^\top H^\top X) = \sum_{j=1}^{k} c_j^\top \sum_{x_i \in C_j} x_i.$$

Thus we deduce that

$$\|X - HC\|_F^2 = \sum_{i=1}^{n} \|x_i\|^2 - 2 \sum_{j=1}^{k} c_j^\top \sum_{x_i \in C_j} x_i + \sum_{j=1}^{k} m_j \|c_j\|^2.$$

Grouping terms cluster-by-cluster and completing the square yields $\sum_{i=1}^{n} \|x_i - c_{r(i)}\|^2$, matching the row-wise result.

Problem 7. (Silhouette Score for k-Means.) Consider a clustering $\{C_1, \ldots, C_k\}$ of n points in $\mathbb{R}^d$. When we write $C_j \backslash \{x_i\}$ we mean the set C_j with the point x_i removed. Recall from the lecture that the *silhouette score* of a point x_i in cluster C_j is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

where

$$a(i) = \frac{1}{|C_j| - 1} \sum_{x \in C_j \backslash \{x_i\}} \|x_i - x\|$$

is the average distance to other points in the same cluster, and

$$b(i) = \min_{q \in \{1,\ldots,k\} \text{ s.t. } q \neq j} \frac{1}{|C_q|} \sum_{x \in C_q} \|x_i - x\|$$

is the average distance to the nearest other cluster; note here that $q \in \{1, \ldots, k\}$ s.t. $q \neq j$ means that q is an index in the set $\{1, \ldots, k\}$ that is not equal to j.

a. Compute the silhouette score for each point in the clustering $C_1 = \{(0,0),(1,0)\}$, $C_2 = \{(5,0),(6,0)\}$.

b. Show that $s(i) \in [-1,1]$ and interpret the meaning of $s(i) \approx 1$, $s(i) \approx 0$, and $s(i) \approx -1$.

c. Explain why maximizing the average silhouette score $\frac{1}{n}\sum_i s(i)$ is a reasonable clustering objective.

Solution.
(a) Computation. For $x_1 = (0,0) \in C_1$:

$$a(1) = \|x_1 - x_2\| = 1, \quad b(1) = \frac{1}{2}\big(\|x_1 - (5,0)\| + \|x_1 - (6,0)\|\big) = \frac{5+6}{2} = 5.5.$$

Thus $s(1) = \frac{5.5-1}{5.5} = \frac{4.5}{5.5} \approx 0.82$.
By symmetry, $s(2) \approx 0.82$, $s(3) \approx 0.82$, $s(4) \approx 0.82$.

(b) Range and interpretation. Since $a(i), b(i) \geq 0$ (both are distances), the numerator $b(i) - a(i)$ satisfies:

$$-a(i) \leq b(i) - a(i) \leq b(i).$$

Dividing by $\max\{a(i), b(i)\}$: if $a(i) \geq b(i)$, then $s(i) = (b(i) - a(i))/a(i) \in [-1, b(i)/a(i)] \subseteq [-1,1]$; if $b(i) \geq a(i)$, then $s(i) = (b(i) - a(i))/b(i) \in [-a(i)/b(i), 1] \subseteq [-1,1]$. Thus $s(i) \in [-1,1]$.

- $s(i) \approx 1$: $b(i) \gg a(i)$, point is well-clustered (far from other clusters, close to own).
- $s(i) \approx 0$: $a(i) \approx b(i)$, point is on the boundary between clusters.
- $s(i) \approx -1$: $a(i) \gg b(i)$, point is misclassified (closer to another cluster).

(c) Average silhouette as objective. High average silhouette means compact, well-separated clusters. Unlike the k-means objective (which only measures compactness), silhouette also rewards separation, making it a more balanced metric for clustering quality.

Problem 8. (Variance Decomposition and k-Means.) For a dataset $X = \{x_1, \ldots, x_n\}$, let $\mu = \frac{1}{n}\sum_{i=1}^n x_i$ be the global mean and $\mu_j = \frac{1}{|C_j|}\sum_{x_i \in C_j} x_i$ be the mean of cluster C_j.

(a) Expand $\|u - v\|^2$ and show that

$$\|u - v\|^2 = \|u - w\|^2 + \|w - v\|^2 + 2(u - w)^\top (w - v).$$

(b) Show that

$$\sum_{i=1}^n \|x_i - \mu\|^2 = \underbrace{\sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2}_{\text{within-cluster variance}} + \underbrace{\sum_{j=1}^k |C_j| \|\mu_j - \mu\|^2}_{\text{between-cluster variance}}.$$

This is interesting for the following reason: the total variance splits into within- and between-cluster parts. Minimizing the k-means objective (the within term) implicitly maximizes the between-cluster scatter at fixed total variance, promoting compact, separated clusters

Solution. Let $\mu = \frac{1}{n}\sum_i x_i$, and $\mu_j = \frac{1}{|C_j|}\sum_{x_i \in C_j} x_i$.
(a) Algebraic proof. Start from the left-hand side and insert w by adding and subtracting it:

$$u - v \;=\; (u - w) + (w - v).$$

Then

$$\|u - v\|^2 \;=\; \|(u - w) + (w - v)\|^2 = \|u - w\|^2 + \|w - v\|^2 + 2(u - w)^\top (w - v),$$

using $\|a + b\|^2 = \|a\|^2 + \|b\|^2 + 2a^\top b$. This proves the identity.
(b) Algebraic identity. Using $\|u - v\|^2 = \|u - w\|^2 + \|w - v\|^2 + 2(u - w)^\top (w - v)$ with $w = \mu_j$ for $x_i \in C_j$,

$$\sum_{x_i \in C_j} \|x_i - \mu\|^2 = \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 + \sum_{x_i \in C_j} \|\mu_j - \mu\|^2 + 2 \sum_{x_i \in C_j} (x_i - \mu_j)^\top (\mu_j - \mu).$$

The cross-term vanishes since $\sum_{x_i \in C_j}(x_i - \mu_j) = 0$. Summing over j yields

$$\sum_{i=1}^{n} \|x_i - \mu\|^2 = \underbrace{\sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2}_{\text{within}} + \underbrace{\sum_{j=1}^{k} |C_j| \|\mu_j - \mu\|^2}_{\text{between}}.$$

(c) Interpretation & Connection to k-means variance decomposition. Total variance splits into within- and between-cluster parts. Minimizing the k-means objective (the within term) implicitly maximizes the between-cluster scatter at fixed total variance, promoting compact, separated clusters.
Let $u = x_i$ be a data point, $w = \mu_j$ its cluster mean, and $v = \mu$ the global (dataset) mean. Summing the identity over all $x_i \in C_j$:

$$\sum_{x_i \in C_j} \|x_i - \mu\|^2 = \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 + \sum_{x_i \in C_j} \|\mu_j - \mu\|^2 + 2 \sum_{x_i \in C_j} (x_i - \mu_j)^\top (\mu_j - \mu).$$

The cross term vanishes because $\sum_{x_i \in C_j}(x_i - \mu_j) = 0$. Hence

$$\sum_{x_i \in C_j} \|x_i - \mu\|^2 = \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 \;+\; |C_j|\, \|\mu_j - \mu\|^2.$$

Summing over all clusters $j = 1, \ldots, k$ yields the standard variance decomposition:

$$\sum_{i=1}^{n} \|x_i - \mu\|^2 = \underbrace{\sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2}_{\text{within-cluster variance}} + \underbrace{\sum_{j=1}^{k} |C_j|\, \|\mu_j - \mu\|^2}_{\text{between-cluster variance}}.$$

Problem 9. (Linear Approximation of a Norm.) Recall that the first order Taylor approximation $\hat{f}$ of a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ at x_0 is

$$\hat{f}(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0).$$

a. Compute the gradient of the ℓ_2 norm $f(x) = \|x\|_2$ at a point x_0 with all nonzero entries.

b. Write the first order Taylor approximation of $f(x) = \|x\|_2$ at a point x_0 with all nonzero entries.

c. Repeat parts (a) and (b) for the case $f(x) = \|x\|_1$.

Solution.

a. The gradient of $f(x) = \|x\|_2$ at a point $x_0 \neq 0$ is

$$\nabla f(x_0) = \frac{x_0}{\|x_0\|_2}.$$

This follows since $\partial f/\partial x_i = x_i/\|x\|_2$ for $x \neq 0$.

b. The first-order Taylor approximation at x_0 is

$$\hat{f}(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0).$$

Substituting,

$$\hat{f}(x) = \|x_0\|_2 + \frac{x_0^\top}{\|x_0\|_2}(x - x_0).$$

c. Consider $f(x) = \|x\|_1 = \sum_{i=1}^d |x_i|$. For **part a** we have that the derivative with respect to coordinate x_i is

$$\frac{\partial f}{\partial x_i}(x) = \operatorname{sign}(x_i) = \begin{cases} +1, & x_i > 0, \\ -1, & x_i < 0. \end{cases}$$

Therefore, when x_0 has all nonzero entries,

$$\nabla f(x_0) = \big(\operatorname{sign}(x_{0,1}), \ldots, \operatorname{sign}(x_{0,d})\big)^\top.$$

For **part b**, the first-order Taylor approximation at x_0 is

$$\hat{f}(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0).$$

Substituting,

$$\hat{f}(x) = \|x_0\|_1 + \sum_{i=1}^d \operatorname{sign}(x_{0,i})(x_i - x_{0,i}).$$

Bonus Problem 1. (Linear Approximation of a Norm.) For Problem 9 above, solve this extra bonus pat. Both the ℓ_1 and the ℓ_2 norm are not differentiable at points with some zero entries. However, we can still write a "subgradient." Consider $f(x) = \|x\|_2$ and the case $x_0 = 0$. For a unit vector v, compute the directional derivative

$$\lim_{t \downarrow 0} \frac{\|tv\|_2 - \|0\|_2}{t}.$$

Does this limit depend on v? What does this tell you about whether a single gradient exists at the origin?

Solution. Let $f(x) = \|x\|_2$ and fix a *unit* vector $v \in \mathbb{R}^d$ (i.e., $\|v\|_2 = 1$). Consider the directional derivative of f at the origin along v:

$$D_v f(0) := \lim_{t \downarrow 0} \frac{f(0 + tv) - f(0)}{t} = \lim_{t \downarrow 0} \frac{\|tv\|_2 - 0}{t} = \lim_{t \downarrow 0} \frac{t\|v\|_2}{t} = \|v\|_2 = 1.$$

Thus, for every unit direction v, the directional derivative exists and equals 1. (For a general nonzero direction w, writing $w = \|w\|_2\, v$ with v unit yields $D_w f(0) = \|w\|_2$.)

Does this imply a gradient exists at 0? No. If a (Euclidean) gradient $\nabla f(0)$ existed, it would have to satisfy the identity

$$D_v f(0) \;=\; \nabla f(0)^\top v \qquad \text{for all unit } v,$$

i.e., the map $v \mapsto D_v f(0)$ would be a linear functional of v. But we have shown $D_v f(0) = 1$ for *every* unit v, which cannot equal $\nabla f(0)^\top v$ for a fixed vector $\nabla f(0)$: pick any unit v orthogonal to $\nabla f(0)$; then $\nabla f(0)^\top v = 0 \neq 1$. This contradiction shows that no single gradient vector can represent all directional derivatives at the origin.
Thus $f(x) = \|x\|_2$ is *not differentiable at the origin* $x = 0$, even though all directional derivatives there exist (and equal 1 along unit directions).

We can also describe the subdifferential at the origin.
At $x_0 = 0$, the ℓ_2 norm is not differentiable (the denominator vanishes). However, we can still describe the subdifferential. The subgradient at 0 is any vector $g \in \mathbb{R}^d$ with $\|g\|_2 \leq 1$. In other words,

$$\partial f(0) = \{g \in \mathbb{R}^d : \|g\|_2 \leq 1\}.$$

Geometrically, the subdifferential at the origin is the closed unit ball in $\mathbb{R}^d$.
At $x_0 = 0$, the expression $\nabla f(x) = x/\|x\|_2$ is undefined, so f is not differentiable at the origin. However, f is convex, so we can describe its *subdifferential* at 0:

$$\partial f(0) = \{g \in \mathbb{R}^d : f(y) \geq f(0) + g^\top (y - 0) \;\; \forall y\}.$$

It can be shown that this set is exactly the closed unit ℓ_2 ball:

$$\partial f(0) = \{g \in \mathbb{R}^d : \|g\|_2 \leq 1\}.$$

Thus:

- For $x_0 \neq 0$, f is differentiable with gradient $\nabla f(x_0) = x_0/\|x_0\|_2$.
- For $x_0 = 0$, f is not differentiable, but every vector g with $\|g\|_2 \leq 1$ is a valid subgradient.

At a point x_0 with some zero entries, the ℓ_1 norm is not differentiable because the absolute value is not differentiable at zero. However, the subgradient of $|x_i|$ at $x_i = 0$ is the interval $[-1, 1]$. Thus, a subgradient g of $f(x) = \|x\|_1$ at such an x_0 can be chosen coordinate-wise as

$$g_i = \begin{cases} \mathrm{sign}(x_{0,i}), & x_{0,i} \neq 0, \\ \theta, & x_{0,i} = 0 \text{ with } \theta \in [-1, 1]. \end{cases}$$

Therefore one valid subgradient vector is obtained by assigning any value in $[-1, 1]$ for each zero coordinate.

Bonus Problem 2. (k-NN with Adversarial Perturbations.) Consider a 1-NN classifier trained on points $\{(x_i, y_i)\}_{i=1}^n$ in $\mathbb{R}^d$.

a. Given a query point x with nearest neighbor x_j, find the *minimum* ℓ_2 perturbation δ such that $x + \delta$ has a different nearest neighbor. Express your answer in terms of the training points and x.

b. Show that if training points are sparse (far apart), the classifier is more robust to perturbations.

c. Design a simple 2D example where a perturbation of magnitude $\|\delta\|_2 = 0.1$ changes the classification from $+1$ to -1.

Solution.

(a) Minimum ℓ_2 perturbation to change the nearest neighbor.

Setup. Let $x \in \mathbb{R}^d$ be the query, x_j its nearest neighbor, and fix any competitor x_k ($k \neq j$). The set where x_j and x_k are *equidistant* is the perpendicular-bisector hyperplane

$$H_{jk} = \Big\{ z : \ \|z - x_j\|_2^2 = \|z - x_k\|_2^2 \Big\} = \Big\{ z : \ (x_k - x_j)^\top z = \tfrac{1}{2}\big(\|x_k\|_2^2 - \|x_j\|_2^2\big) \Big\}.$$

Write

$$n_{jk} := x_k - x_j \neq 0, \qquad d_{jk} := \tfrac{1}{2}\big(\|x_k\|_2^2 - \|x_j\|_2^2\big),$$

so $H_{jk} = \{z : \ n_{jk}^\top z = d_{jk}\}$.

Distance from x to the bisector. The Euclidean distance from x to the hyperplane H_{jk} is

$$\operatorname{dist}(x, H_{jk}) = \frac{|\, n_{jk}^\top x - d_{jk} \,|}{\|n_{jk}\|_2}.$$

Now expand $n_{jk}^\top x - d_{jk}$ in terms of *squared* distances to x_j, x_k:

$$\begin{aligned} n_{jk}^\top x - d_{jk} &= (x_k - x_j)^\top x - \tfrac{1}{2}(\|x_k\|^2 - \|x_j\|^2) \\ &= \tfrac{1}{2}\big(2x_k^\top x - \|x_k\|^2\big) - \tfrac{1}{2}\big(2x_j^\top x - \|x_j\|^2\big) \\ &= \tfrac{1}{2}\big(\|x - x_j\|_2^2 - \|x - x_k\|_2^2\big). \end{aligned}$$

Hence

$$\boxed{\operatorname{dist}(x, H_{jk}) = \frac{\big|\|x - x_j\|_2^2 - \|x - x_k\|_2^2\big|}{2\,\|x_k - x_j\|_2}.}$$

Because x_j is the nearest neighbor of x, we have $\|x - x_j\| \leq \|x - x_k\|$, so the numerator is nonnegative and the absolute value can be dropped:

$$\boxed{\rho_{j\to k}(x) := \operatorname{dist}(x, H_{jk}) = \frac{\|x - x_k\|_2^2 - \|x - x_j\|_2^2}{2\,\|x_k - x_j\|_2}.}$$

Minimal perturbation magnitude (over all competitors). To make *any* other training point beat x_j, the smallest move is to the *closest* bisector among $\{H_{jk} : k \neq j\}$:

$$\boxed{\|\delta^\star\|_2 = \min_{k\neq j} \ \rho_{j\to k}(x) = \min_{k\neq j} \ \frac{\|x - x_k\|_2^2 - \|x - x_j\|_2^2}{2\,\|x_k - x_j\|_2}.}$$

A concrete perturbation vector. For the index k attaining the minimum above, an ℓ_2-shortest perturbation that reaches H_{jk} is the orthogonal projection of x onto H_{jk} along the normal n_{jk}:

$$\boxed{\delta^\star = -\,\frac{n_{jk}^\top x - d_{jk}}{\|n_{jk}\|_2^2}\, n_{jk} = -\,\frac{\|x - x_j\|_2^2 - \|x - x_k\|_2^2}{2\,\|x_k - x_j\|_2^2}\,(x_k - x_j).}$$

Any infinitesimal step beyond H_{jk} in the same direction makes x_k strictly closer than x_j.

Equivalent "angle" (cosine) form (optional). Let $d_j = \|x - x_j\|$, $d_k = \|x - x_k\|$, and let $\theta \in [0, \pi]$ be the angle between n_{jk} and the vector from x to the midpoint $m_{jk} = \frac{1}{2}(x_j + x_k)$. One can show

$$\rho_{j \to k}(x) = \frac{d_k^2 - d_j^2}{2\,\|x_k - x_j\|} = \frac{(d_k - d_j)(d_k + d_j)}{2\,\|n_{jk}\|} = \frac{d_k - d_j}{2\cos\phi},$$

where ϕ is the angle between the direction n_{jk} (the boundary normal) and the line from x to the bisector. This recovers the common geometric intuition: the "gap" shrinks when you move in a direction more aligned with the normal (large $\cos\phi$) and grows as you move tangentially.

(b) Sparse points $\implies$ robustness. If $\|x_k - x_j\|_2$ is large (sparse), then the denominator in $\|\delta\|$ is large, but the numerator grows more slowly. The decision boundary is farther from x, requiring larger perturbations to flip the prediction.

(c) Example. Let $x_1 = (-1, 0)$ with $y_1 = +1$, $x_2 = (1, 0)$ with $y_2 = -1$, and query $x = (0.09, 0)$. Then x is closer to x_2. A perturbation $\delta = (-0.1, 0)$ moves x to $(-0.01, 0)$, flipping the nearest neighbor to x_1.